Class-05 | 07-06-2026 | Shayan (1)
CSE323 | Compiler Design

Ambiguous Grammar

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid ID$

Non-terminal {E, T, F}

Terminal {+, *, (,), ID}

Starting variable {E}

Production Rules

$E \rightarrow E+T$

$E \rightarrow T$

$T \rightarrow T*F$

$T \rightarrow F$

$F \rightarrow (E)$

$F \rightarrow ID$

Example

$A \rightarrow BC \mid d \mid f$

$B \rightarrow g \mid h$

$C \rightarrow e \mid \varepsilon$

Non-terminal: {A, B, C}

Terminal: {BC, d, f, g, h, e, ε}

Starting variable: {A}

Production Rules

$A \rightarrow BC$

$A \rightarrow d$

$A \rightarrow f$

$B \rightarrow g$

$B \rightarrow h$

$C \rightarrow e$

$C \rightarrow \varepsilon$

$E \rightarrow E+E$

$E \rightarrow E*E$

$E \rightarrow (E)$

$E \rightarrow ID$

Starting variable {E}

Terminal {E}

Non terminal {+, *, (,), ID}

Derivation

Input: ID+ID*ID

Left most Derivation

$E \rightarrow E+E$

$\rightarrow ID+E$

$\rightarrow ID+E*E$

$\rightarrow ID+ID*E$

$\rightarrow ID+ID*ID$

$E \rightarrow E*E$

$\rightarrow E+E*E$

$\rightarrow ID+E*E$

$\rightarrow ID+ID*E$

$\rightarrow ID+ID*ID$

Shayan

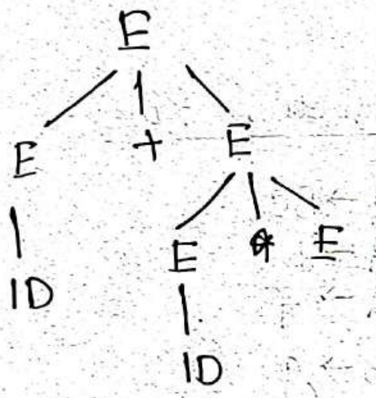
Right most Derivation ①

$E \rightarrow E+E$
$\rightarrow E+E*E$
$\rightarrow E+E*ID$
$\rightarrow E+ID*ID$
$\rightarrow ID+ID*ID$

RMD ②

$E \rightarrow E*E$
$\rightarrow E*ID$
$\rightarrow E+E*ID$
$\rightarrow E+ID*ID$
$\rightarrow ID+ID*ID$

Parse Tree
LMD



Q. Check whether it is Ambiguous? the grammar

Q. Convert Ambiguous to Unambiguous

↗ Left Recursion

$A \rightarrow Aa \mid B$

↗ Right Recursion

$E \rightarrow A+E$

→ Unambiguous

$E \rightarrow E+T$
$E \rightarrow E*T$
$E \rightarrow T$
$E \rightarrow (E)$
$T \rightarrow ID$

$E \rightarrow E+E$
$E \rightarrow E*E$
$E \rightarrow (E)$
$E \rightarrow ID$

$(ID+ID)+ID$

22 Tarikh ল্যাব মিড
15 Tarikh CT

$E \rightarrow E + E$

$E \rightarrow E * E$

$E \rightarrow (E)$

$E \rightarrow ID$

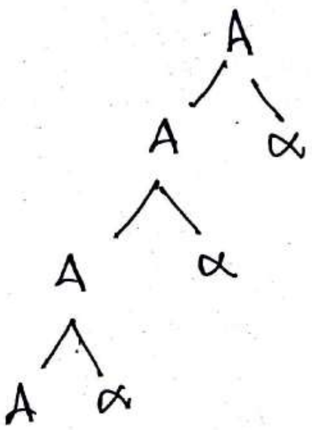
Input: $(ID + ID) * ID$

↗ Left Recursion

```
A(){
    A()  ←
    . . .
    . . .
}
```

eternal rule

$A \rightarrow A\alpha \mid b$



To eliminate the left recursion:

Formula: ← new non-terminal

$A \rightarrow \beta A'$

$A' \rightarrow \alpha A' \mid \varepsilon$

#Example

$\underset{A}{E} \rightarrow \underset{A}{E}\,\underset{\alpha}{\boxed{+T}} \mid \underset{\beta}{T}$

$E \rightarrow TE'$

$E' \rightarrow +TE' \mid \varepsilon$

$\underset{A}{T} \rightarrow \underset{A}{T}\,\underset{\alpha}{\boxed{*F}} \mid \underset{\beta}{F}$

$T \rightarrow FT'$

$T' \rightarrow *FT' \mid \varepsilon$

Shayan

$$\underset{A}{A} \rightarrow \underset{A}{A}\underset{\alpha}{\boxed{Be}} \mid \underset{\beta}{f} \mid \underset{\beta}{g} \mid \underset{A}{A}\underset{\alpha}{\boxed{c}}$$

$$A \rightarrow fA' \mid gA'$$
$$A' \rightarrow BeA' \mid cA' \mid \varepsilon$$

$$\underset{A}{S} \rightarrow \underset{A}{S}\underset{\alpha}{1} \mid \underset{\beta}{a} \mid \underset{\beta}{b} \mid \underset{\beta}{c} \mid \underset{A}{S}\underset{\alpha}{2} \mid \underset{A}{S}\underset{\alpha}{3}$$

$$S \rightarrow aS' \mid bS' \mid cS'$$
$$S' \rightarrow 1S' \mid 2S' \mid 3S' \mid \varepsilon$$

Class - 06 | 08-06-2026 | [signature]
CSE 323 | Compiler Design

Indirect Left Recursion

$S \to Af \mid b$
$A \to Ac \mid Sd \mid Be$
$B \to Ag \mid Sh \mid k$

$A \to A\alpha \mid \beta$ $\quad A \to \beta A'$
$A' \to \alpha A' \mid \varepsilon$

STEP 1

$S \to Af \mid b$
$A \to Ac \mid Afd \mid bd \mid Be$
$B \to Ag \mid Afh \mid bh \mid K$

$A \to Ac \mid (Af \mid b)d \mid Be$
$A \to Ac \mid Afd \mid bd \mid Be$

STEP 2

$S \to Af \mid b$

$A \to \underbrace{A}_{A}\underbrace{c}_{\alpha} \mid \underbrace{A}_{A}\underbrace{fd}_{\alpha} \mid \underbrace{bd}_{\beta} \mid \underbrace{Be}_{\beta}$

$A \to bdA' \mid BeA'$
$A' \to cA' \mid fdA' \mid \varepsilon$

$B \to (bdA' \mid BeA')g \mid (bdA' \mid BeA')fh \mid bh \mid K$
$B \to bdA'g \mid BeA'g \mid bdA'fh \mid BeA'fh \mid bh \mid K$

STEP 3

$S \to Af \mid b$
$A \to bdA' \mid BeA'$
$A' \to cA' \mid fdA' \mid \varepsilon$

$B \to \underbrace{bdA'g}_{\beta} \mid \underbrace{B}_{A}\underbrace{eA'g}_{\alpha} \mid \underbrace{bdA'fh}_{\beta} \mid \underbrace{B}_{A}\underbrace{eA'fh}_{\alpha} \mid \underbrace{bh}_{\beta} \mid \underbrace{k}_{\beta}$

~~$B' \to A'g\,Be \mid eA'fhB \mid bdA'g \mid bdA'fh \mid bh \mid k$~~

$B \to bdA'gB' \mid bdA'fhB' \mid bhB' \mid kB' \mid \varepsilon$
$B' \to \cancel{bdA'gB'}\; eA'gB' \mid eA'fhB' \mid \varepsilon$

Shayan

$X \rightarrow XSb \mid Sa \mid b$

$S \rightarrow Sb \mid Xa \mid a$

STEP 1

$X \rightarrow \underbrace{X}_{A}\underbrace{Sb}_{\alpha} \mid \underbrace{Sa}_{\beta} \mid \underbrace{b}_{\beta}$

$A \rightarrow A\alpha \mid \beta$

$A \rightarrow \beta A'$

$A' \rightarrow \alpha A' \mid \varepsilon$

$X \rightarrow SabX'$

$X' \rightarrow SbX' \mid \varepsilon$

STEP 2

$S \rightarrow Sb \mid (SabX')a \mid a$

$S \rightarrow \underbrace{S}_{A}\underbrace{b}_{\alpha} \mid \underbrace{S}_{A}\underbrace{abX'a}_{\alpha} \mid \underbrace{a}_{\beta}$

$S \rightarrow aS'$

$S' \rightarrow bA' \mid abX'aS' \mid \varepsilon$

Determine whether Left Recursion is direct or indirect (কারণ question এ বলা থাকবে না)

$E \rightarrow E+T \mid T$

$T \rightarrow T*F \mid F$

$F \rightarrow (E) \mid ID$

উপরের নন-Terminal নিচের Line এ use না হলে direct left recursion না হলে indirect recursion হবে।

$\boxed{\text{CT সিলেবাস}}$

1. Error, Types of Errors
2. Left Recursion
3. Left ~~Re~~ Factoring

15 তারিখ CT

↗ Left Factoring

$$A \rightarrow \alpha B_1 \mid \alpha B_2 \mid \alpha B_3 \mid \cdots \mid \gamma_1 \mid \gamma_2 \mid \gamma_3$$

Formula:

$$A \rightarrow \alpha A' \mid \gamma_1 \mid \gamma_2 \mid \gamma_3$$

$$A' \rightarrow B_1 \mid B_2 \mid B_3$$

$$A \rightarrow abcB \mid abcD \mid abcE \mid F \mid G$$

$$abc\underbrace{(B \mid D \mid E)}_{A'} \mid F \mid G$$

$$A \rightarrow abc\,A' \mid F \mid G$$

$$A \rightarrow B \mid D \mid E$$

$$S \rightarrow 101P \mid 10Q \mid R$$

$$\rightarrow \underbrace{10}_{\alpha}\underbrace{(1P \mid Q)}_{A'} \mid \underbrace{R}_{\gamma_1}$$

$$S \rightarrow 10S' \mid R$$

$$S \rightarrow 1P \mid Q$$

$$S \rightarrow abcd \mid abc \mid ab \mid a$$

STEP 1

$$S \rightarrow aS'$$

$$S' \rightarrow bcd \mid bc \mid b \mid \varepsilon$$

STEP 2

$$S \rightarrow aS'$$

$$S' \rightarrow bS'' \mid \varepsilon$$

$$S'' \rightarrow cd \mid c \mid \varepsilon$$

STEP 3

$$S \rightarrow aS'$$

$$S' \rightarrow bS'' \mid \varepsilon$$

$$S'' \rightarrow cd \mid c \mid \varepsilon \rightarrow cS''' \mid \varepsilon$$

$$S''' \rightarrow d \mid \varepsilon$$

Shayan

$S \rightarrow aSSbS \mid aSaSb \mid abb \mid b$